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1. Comparative Analysis of Radiology Report Generation Architectures on the Belarusian Screening Dataset

In our previous Q11 report, we used three Radiology Report Generation (RRG) methods for the Belarusian screening test set:

- lightweight ResNet-18 + LSTM architecture, equipped with pretrained GloVe embeddings (20 million trainable parameters);
- VisionGPT2 architecture (ViT encoder and GPT2 decoder transformers with 210 million trainable parameters);
- fine-tuned multimodal large language model (M-LLM) Qwen3.5-4B (from 33 to 78 million trainable and 4 billion frozen parameters).

1.1. Metrics Description

In this quarter we applied various evaluation metrics to different versions of these three RRG architectures and analyzed the results obtained.

Standard and relatively simple metrics for natural language generation (NLG) like BLEU and ROUGE-L considered as *highly inefficient* for chest X-ray (CXR) reports. However, despite the objective unreliability of evaluating individual messages, these metrics can provide a rough estimation of overall quality. We'll discuss this in the discussion section below.

BLEU and ROUGE-L metrics rely strictly on surface-level or n-gram overlap text processing. In radiology, a model can penalize a perfectly accurate report for using synonyms (e.g., using "fluid collection" instead of "effusion"), or worse, yield a high BLEU score while missing a critical negation (e.g., generating "pneumothorax is seen" instead of "no pneumothorax is seen").

To evaluate automated CXR reports effectively, both for individual messages and overall dataset, the industry has shifted toward information-extraction metrics, composite frameworks, and LLM-as-a-judge error detection methods.

We conducted our evaluation of RRG architectures based on the following metrics, which are ranked below *in decreasing order* of clinical importance and reliability for radiological reports.

Leaders in RRG evaluation.

- GREEN (Generative Radiology Report Evaluation and Error Notation) is the most advanced metric in the field of RRG to date. It was developed by Stanford researchers (Stanford AIMI) in late 2024. It operates on the *LLM-as-a-judge* principle and utilizes a fine-tuned, clinically-oriented language model acting as a judge to identify and classify clinical errors.
- RadCliQ_Inv (Radiology Report Clinical Quality Inverted) is an automated *inverted composite* evaluation metric designed to measure the clinical accuracy and quality of AI-generated radiology reports. By aligning closely with radiologists' expert judgments, it overcomes the limitations of traditional, surface-level language metrics. The suffix "_Inv" means that it serves as an error-counting metric where lower scores indicate higher clinical report quality.
- RaTEScore (Radiological Report Text Evaluation) was explicitly designed to fix RadGraph's flaws. Traditional graph-based metrics like RadGraph often fail to account for medical synonyms and negation expressions. To overcome this, RaTEScore leverages a specialized medical Named Entity Recognition (NER) model to compare the semantic similarity of clinical embeddings based on their relevance.
- RadGraph_F1 (Radiology Graph F1-score) is a specialized NLP metric that evaluates RRG by converting text into a clinical *knowledge graph* and measuring the harmonic mean of accurately matched medical entities and their anatomical relationships.

High diagnostic usefulness. Medical domain-specific semantic metrics. Beyond the leading evaluation frameworks, a significant class of metrics focuses on high diagnostic usefulness through domain-specific semantic alignment:

- Bio_BERTScore_F1 (Bio-Clinical Balanced Score)
- Bio_BERTScore_R (Bio-Clinical Recall)
- Bio_BERTScore_P (Bio-Clinical Precision)
- Bio_Sentence_CosSim (Biomedical Sentence Embedding Similarity)

The prefix "Bio_" denotes their reliance on specialized biomedical language models—such as BioBERT or ClinicalBERT—rather than general-domain NLP counterparts. By incorporating BERT-based architectures, these metrics capture deeper contextual and clinical meanings compared to surface-level token matching (like BLEU or ROUGE). They evaluate text using precision (P), recall (R), balanced F1-scores, or overall sentence embedding similarity (CosSim).

However, they are no longer considered state-of-the-art leaders. While they successfully capture medical semantics at the sentence level, they fundamentally lack the ability to model complex anatomical relationship graphs, explicitly detect clinical negations, or perform high-level error classification like modern LLM-as-a-judge platforms.

Moderate diagnostic usefulness. Cross-domain semantic metrics. Another category consists of cross-domain semantic metrics that evaluate text based on general linguistic similarity rather than specialized medical knowledge. This group comprises:

- BLEURT (Learned Token-to-Sentence Metric)
- BERTScore_F1 (General-Domain Contextual Similarity)
- Semantic_CosSim (General-Domain Sentence Similarity)

Unlike the "Bio_" variants, these metrics rely on models trained on general-domain corpora (such as standard BERT or Wikipedia embeddings). They offer a moderate level of diagnostic usefulness by capturing semantic context and phrasing variations far better than lexical metrics. However, they fundamentally blind to specialized clinical terminology. Consequently, they often fail to differentiate between critical medical conditions and minor diagnostic nuances. Which prevent them from serving as reliable standalone evaluators for medical RRG tasks.

Low diagnostic usefulness. Traditional lexical metrics. The final and most traditional category relies strictly on surface-level text matching, offering the lowest level of diagnostic usefulness for radiology reports. This group includes:

- METEOR (Advanced Lexical Overlap)
- ROUGE-L (Longest Common Subsequence Lexical Overlap)
- BLEU-4 (4-Gram Lexical Precision)

These metrics operate solely on exact token overlap and n-gram precision, completely ignoring semantic context, clinical weights, or medical synonyms. They penalize valid clinical variations and are inherently blind to critical negations. Thus they are widely recognized as inadequate for evaluating the clinical accuracy of RRG systems.

1.2. Code Availability

The complete source code for implementing and reproducing all the text evaluation metrics described above is publicly available. The implementation can be accessed either as interactive Google Colab notebooks via Google Drive or hosted directly on the GitHub repository.

In the form of Google Colab notebooks:

https://drive.google.com/drive/folders/1jcWfnT9MmrB2j2Vt_gvnqluV0AR8aaNy/

On the GitHub repository:

https://github.com/foobar167/junkyard/tree/master/fine_tuning/text_evaluation_metrics

1.3. The results evaluation and analysis

The evaluation results for the Belarusian screening test set are shown in Fig. 1 and 2.

		Leaders in RRG evaluation				High diagnostic usefulness. Domain-specific semantic				Moderate. Cross-domain semantic			Low usefulness		
Model		GREEN	RadCliQ_Inv	RaTEScore	RadGraph_F1	Bio_BERTScore_F1	Bio_BERTScore_R	Bio_BERTScore_P	Bio_Sent_CosSim	BLEURT	BERTScore_F1	Semantic_CosSim	METEOR	ROUGE-L	BLEU-4
Pre-trained Qwen 3.		0.074	0.274	0.308	0.021	0.841	0.849	0.833	0.921	0.551	0.841	0.924	0.130	0.103	0.014
Qwen 3.5 M-LLM	version 01	0.218	0.302	0.554	0.199	0.883	0.866	0.901	0.927	0.624	0.883	0.936	0.199	0.207	0.059
	version 02	0.200	0.298	0.548	0.183	0.883	0.865	0.902	0.927	0.623	0.883	0.933	0.187	0.203	0.052
	version 03	0.217	0.301	0.534	0.163	0.885	0.869	0.901	0.936	0.627	0.885	0.939	0.218	0.219	0.073
	version 04	0.217	0.299	0.528	0.154	0.883	0.867	0.901	0.936	0.625	0.883	0.936	0.208	0.213	0.069
	version 05	0.296	0.315	0.588	0.215	0.895	0.883	0.908	0.949	0.639	0.895	0.948	0.300	0.298	0.115
	version 06	0.294	0.314	0.589	0.217	0.895	0.883	0.908	0.949	0.638	0.895	0.948	0.299	0.297	0.114
Vit + GPT2	ver. 01	0.176	0.295	0.506	0.134	0.871	0.859	0.883	0.926	0.600	0.871	0.922	0.196	0.190	0.061
	ver. 02	0.135	0.284	0.483	0.078	0.863	0.848	0.880	0.906	0.580	0.863	0.907	0.148	0.161	0.037
	ver. 03	0.207	0.307	0.517	0.203	0.879	0.866	0.893	0.937	0.608	0.879	0.928	0.235	0.226	0.086
	ver. 04	0.190	0.302	0.497	0.175	0.871	0.865	0.878	0.937	0.600	0.871	0.928	0.221	0.206	0.086
	ver. 05	0.073	0.286	0.411	0.065	0.862	0.846	0.878	0.909	0.588	0.862	0.914	0.166	0.168	0.047
	ver. 06	0.090	0.288	0.432	0.087	0.868	0.851	0.885	0.919	0.574	0.868	0.922	0.176	0.178	0.050
	ver. 07	0.146	0.295	0.448	0.127	0.869	0.859	0.880	0.929	0.586	0.869	0.924	0.198	0.189	0.070
ResNet-18 + LSTM	top_p=0.1	0.264	0.315	0.561	0.224	0.889	0.883	0.896	0.946	0.628	0.889	0.947	0.299	0.294	0.122
	top_p=0.4	0.260	0.315	0.555	0.220	0.889	0.885	0.893	0.948	0.628	0.889	0.948	0.306	0.294	0.124
	top_p=0.5	0.258	0.315	0.554	0.216	0.889	0.886	0.892	0.949	0.628	0.889	0.948	0.311	0.294	0.124

Fig. 1. Evaluation results with maximum (*orange*) and minimum (*blue*) values

		Leaders in RRG evaluation				High diagnostic usefulness. Domain-specific semantic				Moderate. Cross-domain semantic			Low usefulness		
Model		GREEN	RadCliQ_Inv	RaTEScore	RadGraph_F1	Bio_BERTScore_F1	Bio_BERTScore_R	Bio_BERTScore_P	Bio_Sent_CosSim	BLEURT	BERTScore_F1	Semantic_CosSim	METEOR	ROUGE-L	BLEU-4
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	top_p=0.5	0.258	0.315	0.554	0.216	0.889	0.886	0.892	0.949	0.628	0.889	0.948	0.311	0.294	0.124

Fig. 2. Evaluation results with color map

In Fig. 1 and 2, all values are between 0 and 1 ($[0, 1]$). Larger values are highlighted in orange and smaller values are highlighted in blue. The visual mapping reveals a split in dominance between two distinct architectural approaches:

- Qwen 3.5 M-LLM (version 05) dominates metrics that require tight balancing and strict precision. This demonstrates its superior capability to *limit clinical hallucinations (false positives)* while maintaining a balanced medical profile.
- ResNet-18 + LSTM (top_p=0.1) dominates the foundational semantic alignments. *It achieves the absolute highest score* in RadGraph_F1, Bio_BERTScore_R and matches Qwen 3.5 model on RadCliQ_Inv.

Regarding the VisionGPT2 transformer, versions 5–7 exhibited inferior performance relative to versions 1–4. This discrepancy is attributable to the selection of the visual encoder. We switched from the "vit_base_patch16_224" model for versions 1–4 to "vit_base_patch32_224" for versions 5–7. Experimental results demonstrate that increasing the visual token granularity via 32×32 patches degrade downstream generation quality relative to extracting finer 16×16 patches.

To optimize performance in future iterations, integrating advanced alternative image encoders—such as "swinv2_base_window12to16_192to256_22kft1k" or "convnext_base"—should be considered.

As expected, the baseline Pre-trained Qwen 3.5 variant yielded *the lowest overall scores* across clinical criteria. The clear stratification of the color map in Fig. 2 reveals that optimal performance is tightly clustered within specific optimized architectures, while baseline configurations are heavily penalized by domain-specific metrics.

1.4. Discussion

The quantitative evaluation results in Fig. 1 and 2 reveal a critical divergence between domain-specific and general-purpose metrics, highlighting that metric selection fundamentally impacts the validation of radiology report generation.

Traditional cross-domain semantic metrics (such as Semantic_CosSim and BERTScore_F1) exhibit a severe insensitivity to clinical accuracy; for instance, the baseline Pre-trained Qwen 3.5 model achieves an exceptionally high cross-domain semantic score of 0.924, yet plummets to a near-zero score of 0.074 on GREEN and 0.021 on RadGraph_F1. This sharp contrast empirically validates that surface-level text overlaps and generic embeddings fail to account for medical correctness.

Conversely, advanced frameworks like GREEN, RadCliQ_Inv, and RadGraph_F1 provide the necessary clinical granularity to effectively penalize hallucinations and omissions, demonstrating that a high score on basic language metrics does not guarantee clinical utility.

A closer analysis of the high diagnostic usefulness tier uncovers a distinct architectural trade-off between the fine-tuned large language models and the lightweight recurrent baselines. The fine-tuned Qwen 3.5 M-LLM (version 05) maximizes its precision profile, achieving the top scores in Bio_BERTScore_F1 (0.895) and Bio_BERTScore_P (0.908). This indicates that the large multi-modal framework operates under a conservative generation constraint, prioritizing tightly

bounded and highly precise clinical vocabulary to effectively mitigate false positives. Conversely, the lightweight ResNet-18 + LSTM (top_p=0.1) architecture prioritizes a broader clinical recall, securing the absolute highest score in RadGraph_F1 (0.224) and matching the M-LLM in RadCliQ_Inv (0.315). This recall-heavy dominance suggests that the LSTM framework maximizes entity coverage, making it safer against missing critical acute findings, albeit at the cost of slightly lower precision.

From a computational standpoint, the advanced fine-tuning iterations (versions 05 and 06) demonstrate significant performance gains, which correlate with architectural refinements and optimization scaling. The transition to the fused "adamw_torch_fused" optimizer in versions 05 and 06 stabilizes high-parameter convergence but introduces a massive VRAM overhead. Consequently, versions 05 and 06 are prone to Out-of-Memory (OOM) exceptions on the Tesla V100 GPU architecture, thereby necessitating the use of an NVIDIA H800 GPU. However, this introduces hardware constraints, as the H800 infrastructure is continuously allocated to alternative computational workflows.

The capacity of a lightweight, 20-million parameter ResNet-18 + LSTM model to competitively match or exceed a 4-billion parameter M-LLM on structural metrics like RadGraph_F1 *remains an open research question*. A plausible explanation is that the underlying textual corpus within this specific radiologic screening dataset exhibits low linguistic complexity. While localized lexical synonyms vary, the core semantic structures and diagnostic path patterns are highly uniform and monotonous. This intrinsic dataset simplicity allows a lightweight architecture utilizing pre-trained GloVe embeddings to successfully capture the clinical distribution map without the need for massive parameter scaling.

Finally, because the fine-tuning of the Qwen 3.5 configurations was concluded over a single epoch, extending the training schedule to a second epoch represents a highly promising pathway for further performance optimization.

Comparison Table 1 with the best state-of-the-art (SOTA) analogues.

Table 1 – External contextual benchmark comparison

Metric Group	Evaluation Metric	SOTA World Leader (MIMIC-CXR dataset)	Our Best Model Score (Belarusian Screening dataset)
Clinical Judges	GREEN Score	0.712 (UniRG-CXR-2026) 0.678 (RadAlign-2025)	0.296 (Qwen 3.5 v05)
Graph-Based Alignment	RadGraph_F1	0.485 (MedVersa-2026) 0.418 (Hybrid Graph Net)	0.224 (ResNet-18 + LSTM)
Domain Semantics	Bio_BERTScore_F1	0.912 (EviAgent-2026)	0.895 (Qwen 3.5 v05)
Traditional NLG	BLEU-4	0.185 (IU-Xray SOTA)	0.124 (ResNet-18 + LSTM)

SOTA models use *multi-billion parameter multi-modal agents* trained on massive multi-institutional datasets like MIMIC-CXR and CheXpert. Our current models meet or exceed the *early 2024 baseline standards*, but not the current ones. However, our models are much smaller and definitely cheaper to obtain.

1.5. Conclusions

While the concise format of the Belarusian dataset keeps absolute scores lower than large global benchmarks, our models still generate highly accurate clinical reports. The results prove that fine-tuned M-LLMs offer the best localized precision, but lightweight recurrent baselines can match them in structural alignment when text patterns are highly uniform.